

Channel-Aware Backdoor Attacks Against Federated Infostealer Malware Classification Using Dynamic API-Call and Network Representations

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2026 8th International Workshop on Big Data and Information Security (IWBIS)
16-18th October 2026

Background and Motivation

Infostealer malware

Infostealer malware targets authentication artifacts such as credentials, browser data and session cookies, which can enable account compromise and unauthorized access [1], [2].

Classifying infostealer families therefore matters for threat intelligence, triage and incident response.

Dynamic analysis

Dynamic analysis captures runtime behavior beyond static file properties: API-call sequences represent host-level execution, network artifacts capture communication patterns, and both can be encoded into CNN-compatible images [3], [4], [5].

Federated learning

Federated learning (FL) allows participants to train such a classifier together without moving raw data to a central server, but its reliance on client-submitted updates lets a malicious client inject backdoor behavior [6], [7].

Research Problem and Objectives

Research problem

Most FL backdoor work targets general image or model-level poisoning rather than channel-specific behavior in malware representations [7].

An RGB-stack representation places API-call, network and fused information in separate channels [5], so a trigger in one channel need not behave like a trigger in another.

That behavior remains underexplored on dynamic API-call and network representations, and this paper investigates it in federated infostealer malware classification.

Objectives

- We construct dynamic API-call and network representations from Cuckoo Sandbox reports of five latest and most prevalent infostealer families.
- We select the main representation–backbone pair by comparing RGB-stack and opacity blend with SmallCNN, MobileNetV2 and ResNet18.
- We evaluate channel-aware backdoor attacks in FL using red/API, green/network, blue/fusion and full-RGB triggers, with trigger-control and Multi-Krum defense analysis.

The novelty is to treat the representation's channels as the attack surface: a trigger confined to the fusion channel alone matches one spanning all three.

Overall Research Workflow

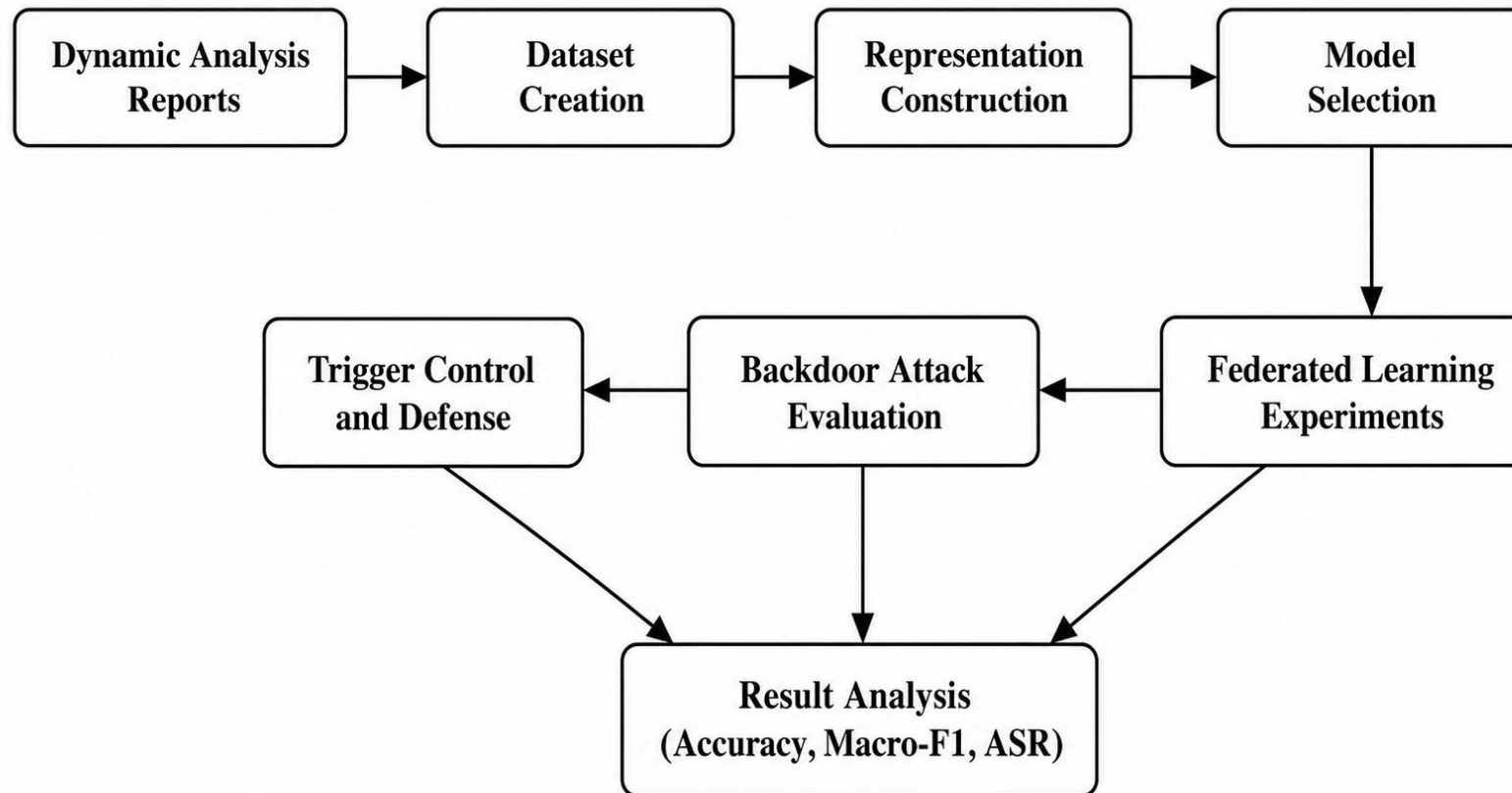


Fig. 1. Overall research workflow of the proposed study.

From Sandbox Reports to Image Tiles Channel-Separated RGB-Stack Representation

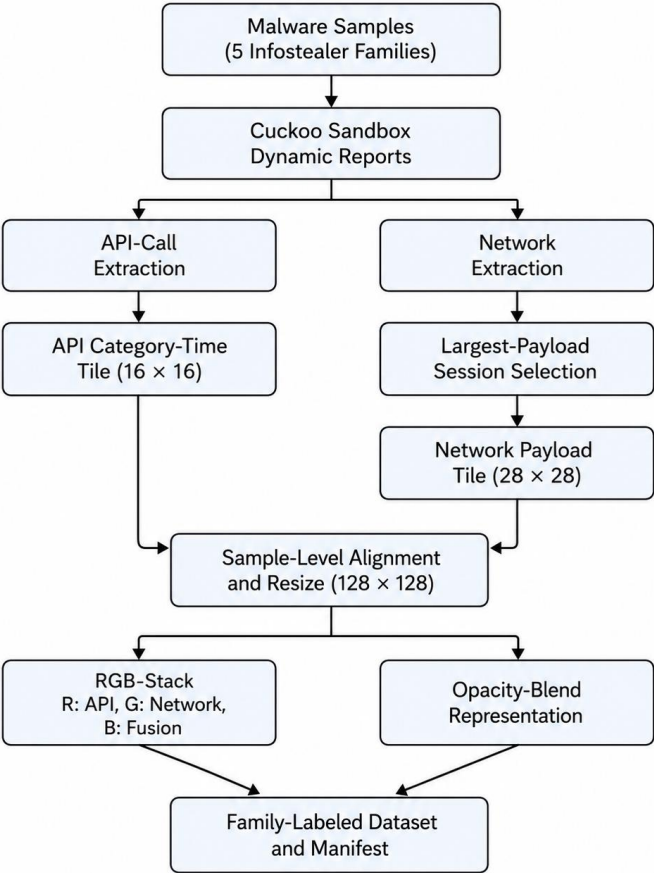


Fig. 2. Dataset creation pipeline from dynamic analysis reports to dynamic API-call and network representations.

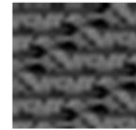
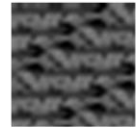
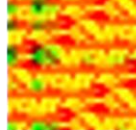
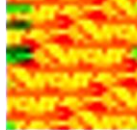
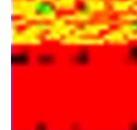
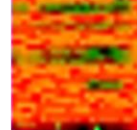
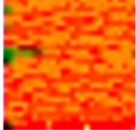
	AgentTesla	FormBook	SalatStealer	StealC	Vidar
API-call sequences Representations	 16x16	 16x16	 16x16	 16x16	 16x16
Network Traffic Representations	 28x28	 28x28	 28x28	 28x28	 28x28
Opacity blend Fusion	 128x128	 128x128	 128x128	 128x128	 128x128
RGB-Stacks Fusion	 128x128	 128x128	 128x128	 128x128	 128x128

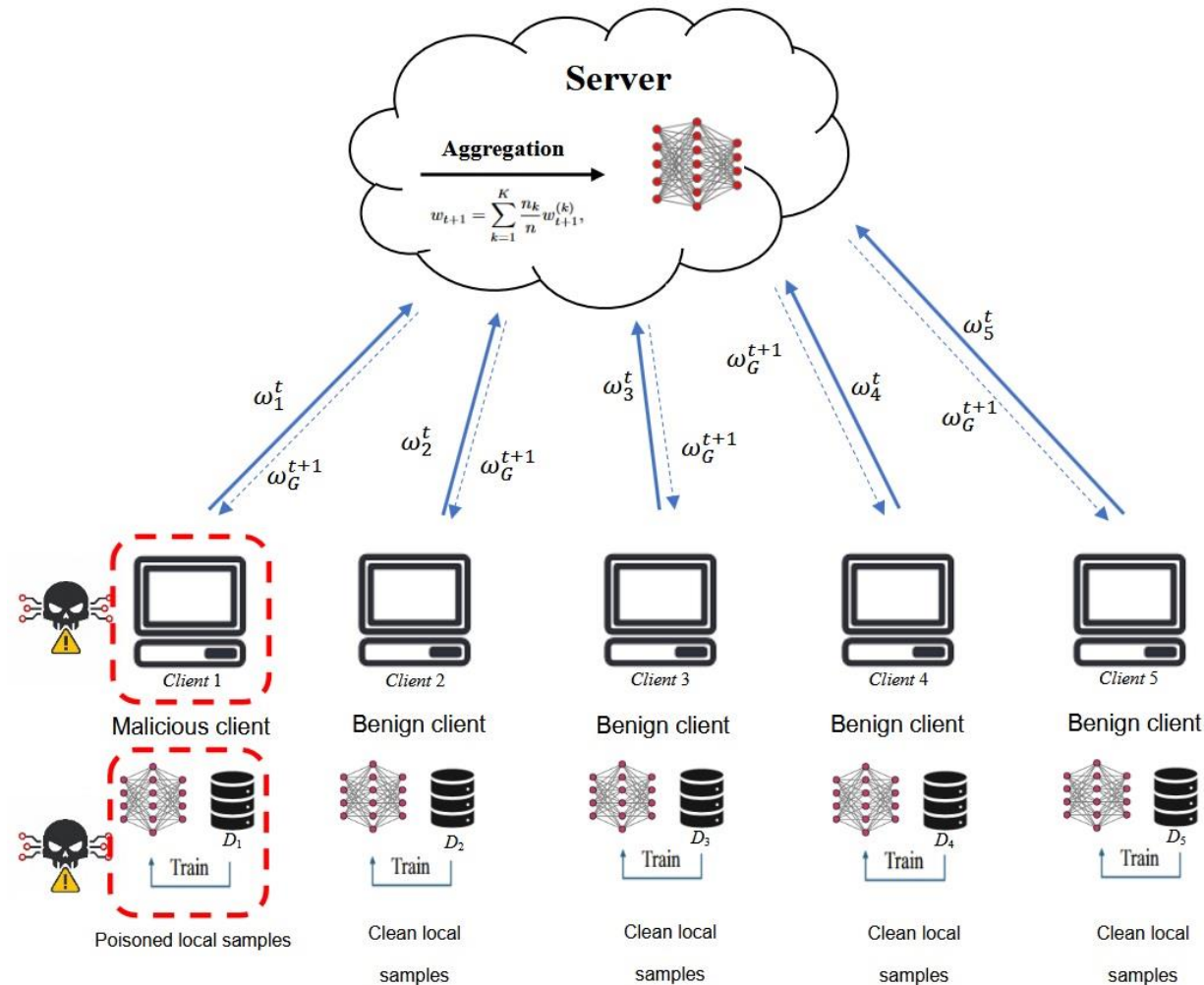
Fig. 3. Representative processed malware representations generated from dynamic API-call and network packet artifacts. Rows illustrate different representation types, while columns correspond to the five infostealer families.

Dataset Creation

TABLE I. Dataset Summary and Stratified Split

Family	Total	Train	Val.	Test
AgentTesla	100	70	15	15
FormBook	100	70	15	15
SalatStealer	100	70	15	15
StealC	100	70	15	15
Vidar	100	70	15	15
Total	500	350	75	75

Channel-Aware Backdoor in Federated Learning



Each client holds 70 images, 14 per family, the partition is balanced and IID.

Targeted backdoor with one malicious client: it inserts trigger into AgentTesla training and relabels them as FormBook [7], [8].

The poison rate is 20% of the attacker's 14 AgentTesla training images, i.e., two images per round, resampled every round.

Because RGB-stack preserves channel-level semantics, four trigger settings are evaluated: red/API, green/network, blue/fusion and full RGB.

Fig. 4. Federated learning architecture used in this study, consisting of one central server and five clients. One client is malicious in the backdoor attack setting.

Training, Defense and Evaluation

TABLE II. Experimental Environment

Component	Specification
Dynamic analysis	Cuckoo Sandbox on Ubuntu
Training environment	Windows, Python / Anaconda
Framework	PyTorch with CUDA 12.8
GPU	NVIDIA GeForce RTX 3080, 12 GB VRAM

Training

Component	Specification
Batch size	16
Learning rates and weight decay	10^{-4}
FL server rounds	50
FL local rounds	2
Optimizer	AdamW

Training Federated Learning

The best model reported FL model is the final-round global model, with no best-validation selection.

Clean accuracy, macro-F1 and ASR are the primary metrics, over seeds 42, 123 and 2026.

Defense and trigger control

Multi-Krum is the primary defense, taken as a standard robust-aggregation baseline [12].

L_2 -norm clipping, coordinate-wise median and trimmed mean are screened alongside it [13].

As a control, the same triggers are applied at test time to the clean FL model.

[12] P. Blanchard et al., NeurIPS, 2017; [13] D. Yin et al., ICML, 2018

Backbone Selection Results

TABLE III. Backbone Selection

Representation	Backbone	Acc.	Macro-F1
Opacity blend	SmallCNN	0.5333	0.5298
Opacity blend	MobileNetV2	0.6933	0.6903
Opacity blend	ResNet18	0.7733	0.7730
RGB-stack	SmallCNN	0.5200	0.5144
RGB-stack	MobileNetV2	0.7200	0.7200
RGB-stack	ResNet18	0.7867	0.7884

Among the RGB-stack backbones in Table III, measured on seed 42, ResNet18 is the strongest, at 78.67% accuracy and 78.84% macro-F1, and is used for all later experiments; the opacity-blend rows are listed for reference only.

Channel-Aware Backdoor Sweep

TABLE IV. Channel-Aware Backdoor Sweep Using Seed 42

Trigger	Clean Acc.	Macro-F1	ASR
Red/API	0.8533	0.8524	0.3333
Green/network	0.8400	0.8419	0.3333
Blue/fusion	0.8400	0.8395	1.0000
Full RGB	0.8400	0.8401	1.0000

Table IV shows that blue/fusion and full-RGB triggers achieve 100% ASR while red/API and green/network do not, and that blue/fusion does so without modifying all three channels.

The difference is categorical. Pooled over the triggered samples, blue/fusion and full RGB differ from their trigger controls with Fisher exact $p = 3.5 \times 10^{-19}$ and $p = 2.6 \times 10^{-18}$, and blue differs from red and green at $p = 4.0 \times 10^{-8}$, whereas red and green are indistinguishable from their own controls (both $p = 1$, 5/15 against 4/15, seed 42 only).

Pooling is legitimate because the split is re-drawn per seed.

Defense Screening

TABLE V. Single-Seed Defense Screening Against Strongest Attacks Using Seed 42

Trigger	Defense	Clean Acc.	Macro-F1	ASR
Blue/fusion	None	0.8400	0.8395	1.0000
Blue/fusion	Clipping	0.8533	0.8550	1.0000
Blue/fusion	Median	0.8133	0.8120	1.0000
Blue/fusion	Trimmed mean	0.8267	0.8244	1.0000
Blue/fusion	Multi-Krum	0.8267	0.8244	1.0000
Full RGB	None	0.8400	0.8401	1.0000
Full RGB	Clipping	0.8533	0.8550	1.0000
Full RGB	Median	0.8400	0.8388	1.0000
Full RGB	Trimmed mean	0.8133	0.8145	1.0000
Full RGB	Multi-Krum	0.7600	0.7542	0.4000

Table V gives a seed-42 screening against both strongest attacks: clipping, coordinate-wise median and trimmed mean all leave ASR at 1.0000.

Only Multi-Krum reduces it, and only under full RGB (1.0000 to 0.4000, clean accuracy 0.8400 to 0.7600), so Multi-Krum is taken to the multi-seed evaluation.

[12] P. Blanchard et al., NeurIPS, 2017; [13] D. Yin et al., ICML, 2018

Multi-Seed Results

TABLE VI. Main Multi-Seed Results Under IID Federated Setting, With Per-Seed Values

Scenario	Channel	Clean Acc.	Macro-F1	ASR / Target Rate	Per seed (42, 123, 2026)
Clean FL	–	0.8267 ± 0.0134	0.8255 ± 0.0104	–	–
Trigger control	Blue/fusion	0.8267 ± 0.0134	0.8255 ± 0.0104	0.1333 ± 0.1155	4/15, 1/15, 1/15
Trigger control	Full RGB	0.8267 ± 0.0134	0.8255 ± 0.0104	0.1556 ± 0.1018	4/15, 1/15, 2/15
Backdoor, FedAvg	Blue/fusion	0.8311 ± 0.0539	0.8320 ± 0.0539	1.0000 ± 0.0000	15/15, 15/15, 15/15
Backdoor, FedAvg	Full RGB	0.8267 ± 0.0611	0.8262 ± 0.0621	1.0000 ± 0.0000	15/15, 15/15, 15/15
Backdoor, Multi-Krum	Blue/fusion	0.8178 ± 0.0154	0.8184 ± 0.0135	0.5111 ± 0.4286†	15/15, 3/15, 5/15
Backdoor, Multi-Krum	Full RGB	0.7956 ± 0.0407	0.7931 ± 0.0427	0.6000 ± 0.3464†	6/15, 15/15, 6/15

† Across-seed ASR range ≥ 0.5: the outcome is bimodal, so the mean describes no individual run and the per-seed counts (out of 15 source samples) should be read instead.
One deviation must be stated: the seed-42 clean baseline in Table VI and the seed-42 trigger controls come from a 30-round, one-local-epoch run. No ASR result is affected; every backdoor and defense run uses the common budget.

Across three seeds, blue/fusion and full-RGB backdoors achieve 100% ASR while keeping clean performance comparable to the clean FL baseline.

In seven of the eight trigger controls the target rate is identical with and without the trigger, so the ASR is not an artifact of the pattern.

Clean Macro-F1 and ASR over FL Rounds

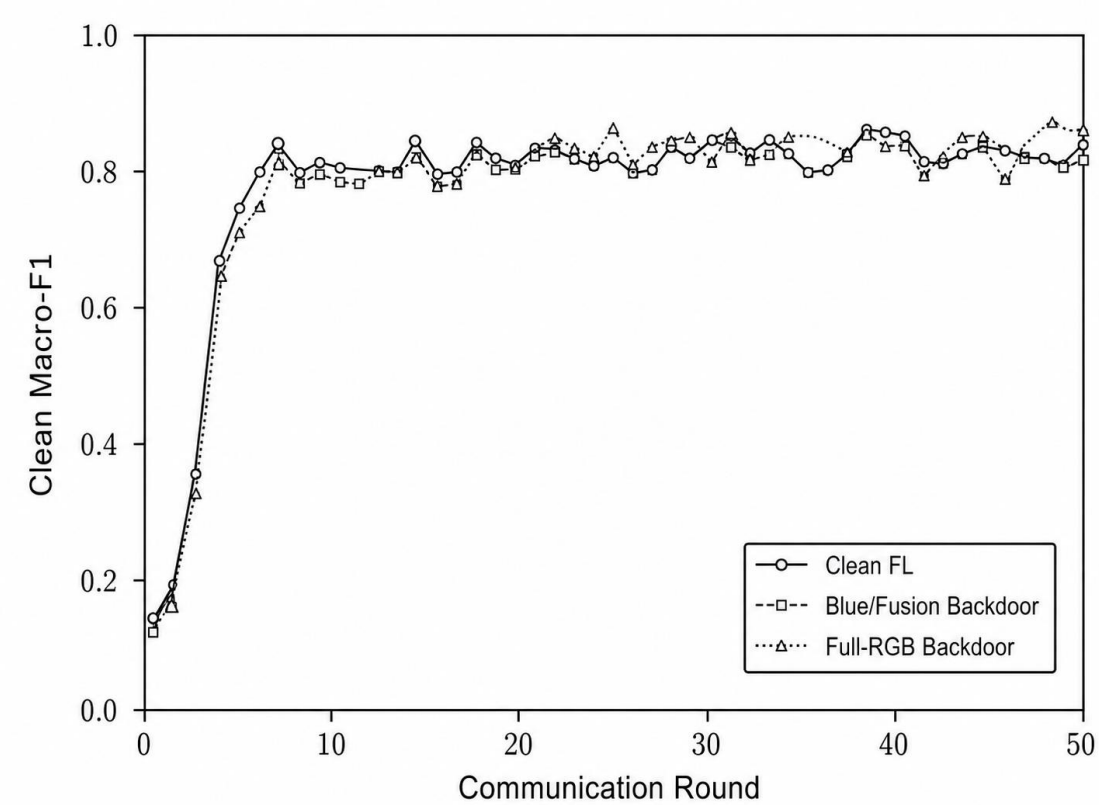


Fig. 5. Clean macro-F1 over FL communication rounds for the clean FL baseline and the strongest backdoor settings, using seed 42.

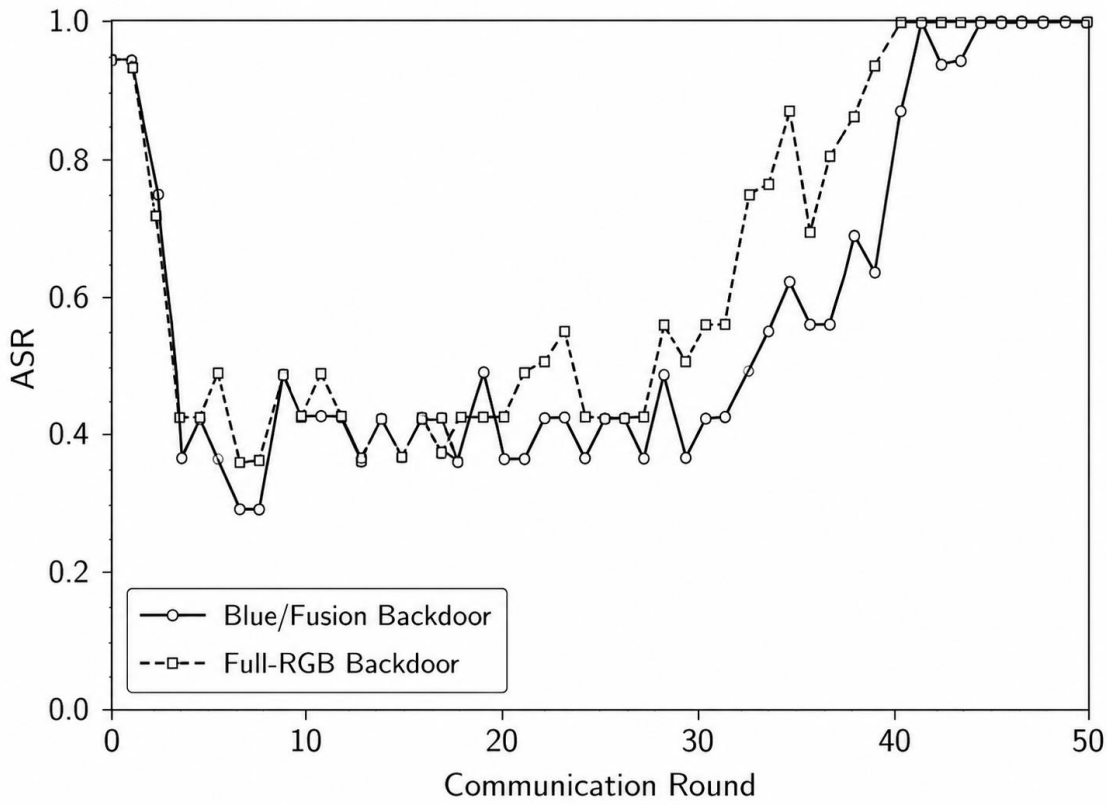


Fig. 6. Attack success rate (ASR) over FL communication rounds for the strongest blue/fusion and full-RGB backdoor settings, using seed 42.

Defense Analysis: Bimodal Rather Than Partial

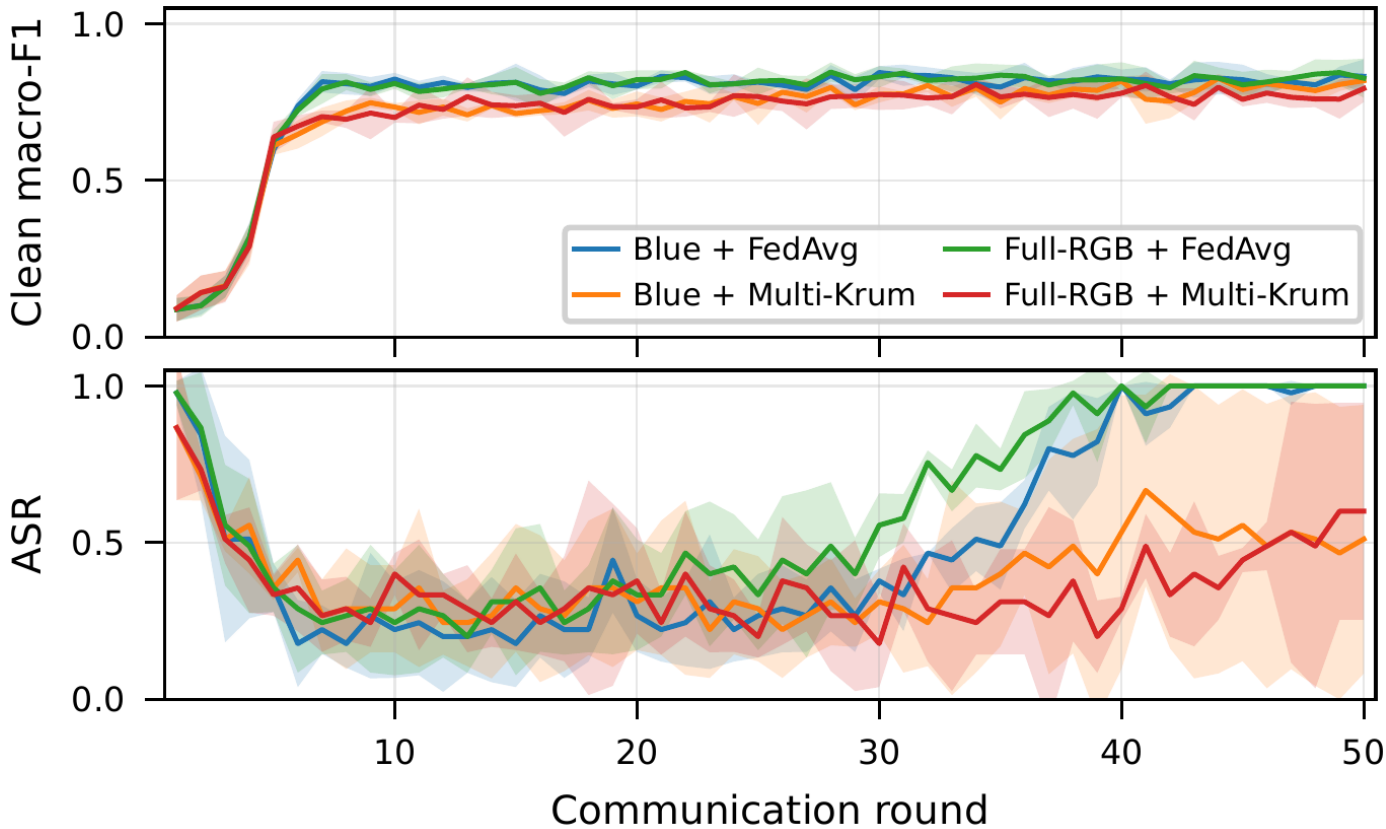


Fig. 7. Per-round clean macro-F1 and ASR across three random seeds for the blue/fusion and full-RGB backdoor settings under FedAvg and Multi-Krum. Shaded areas denote standard deviation across seeds.

The last two rows of Table VI report Multi-Krum per seed as well as on average, because the mean is misleading here.

For both triggers the across-seed ASR range is at least 0.5, with nothing in between: on some seeds the backdoor remains fully effective, on the others most of it is suppressed.

The outcome tracks how often the poisoned update survives selection: across the six runs the malicious client is retained in 8–50% of rounds, and that rate correlates with the final ASR ($r = 0.893$, $p = 0.017$, $n = 6$).

Consistent with Table VI, Multi-Krum does not settle at a stable partial suppression level: for each trigger one seed shows no suppression at all, so it is unreliable as a standalone defense.

The Blue/Fusion Channel

The ordering in Table IV

The ordering follows the intensity of the clean images: over all 500 images the mean value inside the trigger region is 214.6 in R, 48.3 in G and 21.6 in B, and 37.7% of red trigger-region pixels are already at least 254, where the trigger changes nothing.

Contrast explains red's failure but not green's, whose contrast is 207.

What it suggests, and what it does not

Since blue is a deterministic edge map of the other two, its effectiveness cannot come from information they lack; what distinguishes it is that it is almost empty, so the trigger is the only strong response there.

That the blue trigger works because its channel is nearly empty, rather than because of what that channel encodes, is consistent with the measurements but not established.

Conclusion and Future Work

- **The novelty is to treat the representation's channels as the attack surface: a trigger confined to the fusion channel alone matches one spanning all three, even though that channel is derived from the other two.**
- Blue/fusion and full-RGB triggers reached 100% ASR across three seeds while keeping clean performance close to the baseline, and in seven of the eight trigger controls the target rate was identical with and without the trigger, so the effect came from poisoning rather than the trigger pattern.
- Multi-Krum reduced ASR bimodally rather than partially, leaving the backdoor fully effective (15/15) on one of the three seeds for each trigger.
- **Channel-aware backdoors therefore pose a serious threat to federated malware classifiers in this controlled IID setting, and motivate representation-aware defenses.**

Limitations and future work

The main limitations are the IID-only partition, a single source–target pair, and three seeds, with the red and green results resting on seed 42 alone. That the blue trigger works because its channel is nearly empty, rather than because of what that channel encodes, is consistent with the measurements but not established; a non-IID partition, a second source–target pair and a contrast-matched trigger are left to future work.

Thank you

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Notes on Tables I, II and VI and Figs. 5–6

Table I

slide 6

Tiles are aligned per sample and recorded in a manifest organized by family, which supports stratified splitting and reproducibility.

With 100 samples per family the dataset holds 500 representations, split 70%–15%–15% into training, validation and test partitions, stratified by family.

Table II

slide 8

Dynamic analysis and model training run on the separate environments listed in Table II.

Table VI

slide 12

One deviation must be stated: the seed-42 clean baseline in Table VI and the seed-42 trigger controls come from a 30-round, one-local-epoch run. Re-training it under the common budget gives 0.7867 accuracy, 0.7869 macro-F1 and a control rate of 6/15 rather than 4/15, so the reported run sets a higher bar for the clean-performance claim and a lower one for the trigger-control argument. No ASR result is affected; every backdoor and defense run uses the common budget.

Figs. 5–6

slide 13

Figures 5 and 6 show clean macro-F1 holding steady while ASR climbs to 100%. The first rounds are not meaningful: the global model is still close to initialization and collapses onto one or two classes, producing the early spike before clean accuracy rises.

The dataset is built from Cuckoo Sandbox reports for five Windows infostealer families: AgentTesla, FormBook, SalatStealer, StealC and Vidar.

API-call events are converted into a 16×16 category-by-time tile by normalizing the execution timeline, mapping API names into predefined behavior categories, and counting category occurrences over fixed time intervals.

To reduce sample-level leakage, only the session with the largest payload is selected for each malware sample, then truncated or padded and reshaped into a 28×28 network tile.

In RGB-stack the API and network tiles are converted to grayscale, min–max normalized per image and resized to 128×128 ; red carries API-call behavior, green network behavior, and blue is the edge map of their average, $B = \text{FindEdges}((R+G)/2)$.

Blue is thus a deterministic function of the other two channels, adds no independent information, and is by far the emptiest: 62.8% of its pixels are exactly zero.

RGB-stack is used here because only it separates the sources by channel.

The selected representation–backbone pair is trained in a simulated FL setting with one central server and five clients. Each client holds 70 images, 14 per family, so the partition is balanced and IID.

We assume a targeted backdoor with one malicious client: it inserts a square image-level trigger into AgentTesla training images and relabels them as FormBook.

The poison rate is 20% of the attacker's 14 AgentTesla training images, i.e., two images per round, resampled every round.

Because RGB-stack preserves channel-level semantics, four trigger settings are evaluated: red/API, green/network, blue/fusion and full RGB.